PART 1 — RULES DERIVED FROM THE DATA

Everything in Part 1 needs a **YES / NO / CORRECTION** from the engineer.

1.1 Weight formulas

These reproduce exactly across the corpus. Confirm all three.

#	Applies to	Formula	Status
W1	Fabric components (panels, spouts, baffle, skirt, top, bottom)	weight_g = GSM × width_cm × cut_cm × qty / 10000	Exact on every fabric row tested
W2	Tape / loop / tie / cord / DDP	weight_g = GPM × cut_cm × qty / 100	Exact. GPM = grams per metre (linear), **not** g/m²
W3	LDPE liner	effective_gsm = micron × 1.84 then apply W1	Derivation: 2 tube walls × LDPE density 0.92 g/cm³

CONFIRM: Is W3 correct, and is it always both walls? Does it change for form-fit vs lay-flat vs gusseted liners?

1.2 Geometry — U-PANEL (U+2)

All dimensions computed from **EXTERNAL** L × W × H. n = 174 sheets.

Component	Derived rule	Agreement	Notes
Body panel — width	W + 8	**163/174 = 94%**	Next values: 9.0 (×3), 47.0 (×3), 4.0, 10.0
Body panel — cut length	2H + L + 10	**138/174 = 79%**	+8 on 12 sheets (7%), +9 on 6, +3 on 6, +5 on 3, +11 on 2, +6 on 1
Body panel — qty	1	—	U-piece forms bottom + 2 walls
Side panel — width	L + 8	**156/166 = 94%**	Next: 9.0 (×3), 10.0, 14.0

Component	Derived rule	Agreement	Notes
Side panel — cut length	H + 8	**154/166 = 98%**	Next: 7.0 (×6), 10.0, 11.0
Side panel — qty	2	**100%** (n=30 resolvable)	
Bottom panel	**NONE**	—	U-piece forms the bottom. Do not add one.
Reinforcement fabric	**NONE**	—	RF is for cross-corner only
Top panel (when fitted)	(L + 10) × (W + 10)	48% mode (n=23)	+8 and +12 also appear

The +8 / +10 conflict, broken down by customer:

Customer	n	Body-cut allowance values observed
EXPORT ENQUIRY	31	+10 ×29, +12.4 ×1, +25 ×1
PO 28	18	+10 ×18
TECHNOPACK	12	+10 ×11, +8 ×1
EXPORT INQUIRY	11	+9 ×6, +10 ×2, −28 ×3
DOMESTIC	8	+3 ×4, +8 ×3, +6 ×1
JUMBO BAG SPEC	7	+10 ×7
DOMESTIC INQUIRY	5	+10 ×4, +11 ×1
AIRSON	4	+10 ×4
NUDITEX	4	+10 ×4
VIJAYNAGAR	4	+8 ×3, +10 ×1

NOTE: The reference sheet supplied (90X90X90 CM.xlsx, contents 85×85×85) uses **+8**: cut 263 = 2(85) + 85 + 8. It is in the 7% minority. See question **B1**.

1.3 Geometry — CIRCULAR / TUBULAR

n = 104–113 sheets.

Component	Derived rule	Agreement	Notes
Body — flat woven width	W + L (no seam allowance)	mode 38% (n=104)	Comes off the loom as a tube. On 90×90 bags, width − W = 90, i.e. width = 180 = W+L
Body — cut length	H + 8	**52%** (n=104)	+10 is the next most common, then +7, +6
Body — qty	2 (two tube walls)	—	Body implied qty = 2 on 39% of all sheets, matching the circular share
Bottom panel	(L + 10) × (W + 10)	**86%** (n=94)	Next: +12, +30, +13
Top panel	(L + 10) × (W + 10)	**81–82%** (n=68)	Next: +12, +30, +13
Side panel	**NONE**	—	Circular has no side panels

1.4 Geometry — 4-PANEL BAFFLE (Q-BAG)

n = 70 sheets.

Component	Derived rule	Agreement	Notes
Body panel — width	W + 8	**64%** (n=70)	Next: +7, +12, −12
Body panel — cut length	H + 8	**57%** (n=70)	Next: +7, +10
Body panel — qty	4	—	Body implied qty = 4 on 18% of all sheets
Bottom panel	(L + 8) × (W + 8)	**61% / 59%** (n=70)	Next: +7, +12, +6
Top panel	(L + 8) × (W + 8)	**64%** (n=58)	Next: +7, +13, +12

Component	Derived rule	Agreement	Notes
Baffle — qty	4	**100%** (n=53)	
Baffle — GSM	typically 130, both-side laminated	—	

Baffle fabric marking note found verbatim on a production sheet:

"BAFFLE FABRIC 108 CM WIDTH, FROM CORNER 38 CM BAFFLE MARK AND BAFFLE TO BAFFLE MARK DISTANCE 32 CM AND LAST BAFFLE MARK TO CORNER DISTANCE 38 CM, ON BAFFLE MARK DOUBLE WARP 2.5 CM. REST ALL FABRIC WITHOUT MARK"

Another instruction line, ×6:

"4 BAFFLE WITH 80+20 GSM ONE SIDE LAMINATED 10 HOLES AND STITCHING FROM BOTTOM AND TOP 10 CM"

1.5 Geometry — 4-PANEL (no baffle)

n = 11 sheets. **Too few to be reliable — treat as provisional.**

Component	Derived rule	Agreement
Body panel — width	W + 8	55% (n=11)
Body panel — cut length	H + 7	55% (n=11) — +8 next
Bottom panel	(L + 8) × (W + 8)	50% / 38% (n=8)
Top panel	(L + 8.5) × (W + 8.5)	30% (n=10) — +8, +10, +7.5 also

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1.6 Lifting loops — SOLVED

n = 226 testable sheets.

```
loop_cut_cm = 2 × free_height + LL + SL
loop_qty    = 4
loop_weight = GPM × (cut_cm / 100) × 4
```

Agreement: 197 / 226 = 87.2% exact match.

File	Sheet	qty	FH	LL	SL	Actual	Formula
DOMESTIC ENQUIRY	SAHYADRI STARCH	4	30	90	55	210	205
DOMESTIC ENQUIRY	HONDA BAG WITH L	2	122	96	96	330	436
DOMESTIC ENQUIRY	HONDA BAG WITHOUT	2	122	96	96	362	436
DOMESTIC ENQUIRY	MADHU SILICA	4	25	85	40	176	175

Most common residual when it misses (cut − LL − SL): 65 cm (×6), 51 (×3), 55 (×3), 135 (×2), 170 (×2).

please go with Actual calculation

1.7 Quantity — it is never written down

There is no QTY column anywhere in the sheet. Quantity is only recoverable by dividing the recorded weight by the unit weight implied by the geometry. This is the most fragile step in reading these sheets.

Component	n	Implied quantity distribution
BODY PANEL	394	qty1 **43%** · qty2 **39%** · qty4 **18%**
SIDE PANEL	30	qty2 **100%**
BOTTOM PANNEL	106	qty1 **97%** · qty2 3%
BOTTOM PANEL	97	qty1 **82%** · qty2 16% · qty4 1%
TOP PANNEL	131	qty1 **99%** · qty2 1%
TOP PANEL	65	qty1 **100%**
LOOP	363	qty4 **99%** · qty2 1%
BAFFLE	53	qty4 **100%**
BAFFLE FABRIC	8	qty4 **100%**
FS TIE	77	qty1 **94%** · qty2 6%
DS TIE	70	qty1 **100%**

qty in double layer bag

Same thing

Quantity as per customer required

Component	n	Implied quantity distribution
TIE/IRIS ROPE	57	qty1 **100%**
TIE	68	qty1 87% · qty2 10% · qty28 3%
DDP	52	**qty100 92%** · qty10 4% · qty96 2% · qty1 2% ← **ANOMALY, see E3**
FILLING SPOUT	95	qty1 **100%**
D/S	98	qty1 **100%**

Interpretation: body qty1 = U-panel, qty2 = circular (two tube walls), qty4 = 4-panel. **Quantity is a function of family.**

DDP As per Drawing MOQ

1 pcs

1.8 Lamination notation

Notation	Count	Meaning (to confirm)
70+20 LAMINATED	184	base fabric 70 + coating 20 → **GSM column reads 90 (the total)**
LAMI 65+25	25	= 90 total
60+20 GSM	25	= 80 total
145+30 LAMINATED	22	= 175 total
180+30 LAMINATED	20	= 210
65+20 LAMINATED	16	= 85
170+30 LAMINATED	14	= 200
80+20 LAMINATED	14	= 100
70+20 GSM	14	
60+20 LAMINATED	14	
175+30 UNLAMINATED	**12**	**Contradictory — says a lamination figure then "UNLAMINATED". See C8**

Lamination (Breakup)

1. ~~50 - 65 GSM - 20 coating~~
2. ~~66 - 110 GSM - 25~~

GSM / Lamination

1. 50 - 65 => 15 coating
2. 66 - 130 => 20 coating
3. 130 - 150 => 25 coating
4. 151 - 210 => 30 coating
5. 211 - 240 => 35 coating

Lamination As per Above if else mentioned in PO anything please go with this.

Notation	Count	Meaning (to confirm)
100+20 LAMINATED	11	
120+30 LAMINATED	10	
90+20 LAMINATED	10	
150+30 LAMINATED	10	
65+20	9	
15+90+15	**9**	**Both-side lamination? 15 + 90 + 15 = 120. See C8**
140+25GSM	9	
LAMI 65+20	8	
140+30 BLACK LAMINATION	8	
70+25 LAMINATED	8	
55+15 LAMINATED	8	
65+25	8	
70+20	7	
60+20	6	

1.9 Target bag weight is a CONSTRAINT, not an output

n = 350 sheets stating both a bag weight and a grand total.

Metric	Value
Median gap between stated weight and component total	**0.67%**
Within 1%	**61%**
Within 2%	**78%**

Target Bag weight ≥ PO Bag weight minus 1%.

Header notation: the loop line reads 4X25 · LL 60 CM · SL 40 CM · white · 50 MM/WHITE

- 4X25 = **quantity × free height (cm)**
- LL = long attachment length
- SL = short attachment length

okay

Worked confirmations:

Sheet	qty × FH	LL	SL	Formula gives	Actual cut	Weight check
85×85×85 reference	4×25	65	35	150	**150** ✓	35 g/m × 1.50 m × 4 = **210 g** ✓ okay
DOMESTIC BAFFLE	4×25	60	40	150	**150** ✓	45 × 1.50 × 4 = **270 g** ✓ okay
CONDUCTIVE BAG	4×25	30	30	110	**110** ✓	50 × 1.10 × 4 = **220 g** ✓ okay
EXPORT BAG TYPE-1	4×30	95	40	195	**195** ✓	40 × 1.95 × 4 = **312 g** ✓ okay
EXPORT BAG TYPE-2	4×30	95	50	205	**205** ✓	40 × 2.05 × 4 = **328 g** ✓ okay

Loop quantity: 4 on **358 / 363 = 99%** of sheets. qty 2 on 5.

Observed ranges: free height 22–122 cm, LL 30–140 cm, SL 30–96 cm.

Free Height × 2 + 65% of Bag Height LL + 35% of Bag Height of SL

IMPORTANT: The earlier system used LL + SL + 50 and scored 31% within 2 cm. That was simply the wrong formula. Corner-loop geometry was never the hard problem it appeared to be.

The 29 misses (13%), for the engineer to explain:

File	Sheet	qty	FH	LL	SL	Actual	Formula
105X105X120 CM BAFFLE BAG	105X105X120 CM B	4	30	95	65	240	220 okay
ASHAPURA CHAINA CLAY	100X100X200 CM B	4	22	140	730	254	914 okay
ASHAPURA CHAINA CLAY	100X100X200 CM U	4	22	140	80	274	264 okay
ASHAPURA CHAINA CLAY	106X106X115 CM	4	25	80	40	255	170 okay

please go with Actual

Metric	Value
Within 5%	**91%**
Within 10%	**94%**

Tolerance stated verbatim on 7 sheets: BAG WT +/-2 %

GSM predicted by inverting the weight formula against the target

Leave-one-out test, u_panel, n ≈ 279, snapping the prediction to the observed GSM catalogue:

Predictor	Median err	Exact	Within 5%	Within 10%
Load-chart cell (family, SWL, SF)	3.33%	44.7%	64.7%	71.3%
Same customer + same family	3.08%	45.5%	57.9%	—
(target_weight − non_panel_mass) ÷ panel_area	**0.00%**	**55.6%**	**81.0%**	**89.2%**

The panel-area formula itself is accurate to **median 0.39%** against what was actually cut (94.7% within 5%).

This says the engineer works backward from the required weight, using the load chart only as a floor. See questions **C2** and **G3**.

1.10 Stitching / sewing yarn

Metric	Value
yarn_g ÷ external_H — median	**0.467**
p10	0.326
p90	0.721
n	387

The commonly-cited rule of thumb is 0.45 × H, which sits inside this band — but the spread is more than 2×, and the values actually used are **round numbers**, not calculated ones:

Approx yarn cut cal.
(1) Uf 2 pannel ⇒ 0.45 X Bag Height uf
(2) 4-pannel ⇒ 0.65 X Bag H.
(3) 4-pannel Baffle ⇒ 0.70 X Bag H.
(4) Circular ⇒ 0.50 X Bag H.
(5) Single rep ⇒ 0.25 X Bag H.
(6) Two rep ⇒ 0.30 X Bag H.
(7) container Bag (40 feet) ⇒ 300 qrcy
(20 feet) ⇒ 150 qrcy

Yarn weight (g)	22	25	28	30	32	35	38	39	40	42	45	48	50	52	55	60	65	70
Count	6	12	3	15	5	7	2	1	16	15	25	8	13	3	**45**	**62**	**35**	18

This looks **chosen, not calculated**. See question **F1**.

1.11 Body GSM catalogue actually woven, by family

These are the only values that appear. A generated BOM must snap to this list.

Family	n	GSM values observed (× count)
u_panel	174	75×1, 120×8, **125×35**, 130×7, 135×4, 140×16, 145×4, **150×17**, 155×6, **160×23**, 165×7, 170×2, 180×6, 185×4, 190×3, 195×6, 200×9, 210×4, 220×1, 225×1, 230×2, 240×2, 250×3, 260×2, 370×1
circular	113	49×1, 65×2, 71×1, 90×4, 100×1, 103×1, 105×3, 135×2, 143×1, 145×1, 150×3, 155×2, 160×7, 165×8, 170×9, 175×5, 180×3, 185×6, 190×7, 195×9, **200×10**, 205×6, 210×8, 220×2, 230×4, 240×4, 245×3
baffle	70	135×1, 155×3, 160×10, 165×6, 170×9, 175×7, **180×12**, 185×2, 190×4, 200×5, 210×5, 230×6
four_panel	11	160×1, 165×1, 180×5, 190×1, 195×2, 210×1

1.12 Loop belt — width × GPM combinations observed

Belt width	GPM values used (× count)
50 MM	45×41, 40×37, 30×33, 35×20, 28×16, 32×16, 50×11, 42×10 — **eight gramages at one width**
60 MM	40×16
70 MM	55×6, 50×3, 40×6

Uf2 panel → 40-50mm

Corner loop → 40-50mm

Cross corner → 70mm-100mm

Belt width	GPM values used (× count)
72 MM	42×4
75 MM	50×14, 45×10
80 MM	45×15, 55×14, 50×13, 40×7

1.13 The same specification produced different fabric

Taking every group of sheets sharing **family + exact external dimensions + SWL + safety factor**.

- Spec groups with more than one sheet: **108**
- Groups where the chosen body GSM **differs**: **24 (22%)**

Specification	Body GSM chosen	Who ordered it
u_panel 90×90×110 · SWL 1000 · 5:1	**125 · 150 · 210**	AIRSON / AAA / DOMESTIC INQUIRY
u_panel 90×90×120 · SWL 1000 · 5:1	**125 · 130 · 150**	PO 28 / EXPORT ENQUIRY
u_panel 90×90×100 · SWL 1000 · 5:1	**125 · 140**	SPAIN export vs DOMESTIC INQUIRY
u_panel 95×95×110 · SWL 1250 · 5:1	**140 · 145 · 150**	EXPORT BAG TYPE 1 / TYPE 4
u_panel 85×85×90 · SWL 1000	**140 · 165**	U+2 PANNEL sheets 1 and 2
baffle 105×105×135 · SWL 1000 · 5:1	**175 · 230**	PUNEET SIR vs PRINTING
baffle 110×110×125 · SWL 850 · 5:1	**175 · 200**	SAHYADRI STARCH vs KAYADARI
circular 85×85×129 · SWL 1250 · 5:1	**165 · 170 · 190**	VINAYKA / VNK / VM RAJASTHAN
circular 95×95×134	**135 · 200 · 210**	DEV SIR / DG KIM-1 / PRINTING

Specification	Body GSM chosen	Who ordered it
circular 90×90×114	**205 · 230**	SOUTH-1 / SOUTH-2

125 to 210 GSM is a 68% spread on an identical specification. See question **C1**.

1.14 The company's own GSM charts (GSM- GPM VS SWL-SF.xlsx)

Transcribed as-is for confirmation. **Note: there is NO U-PANEL table in this file.**

Two-panel (C-type) cross corner loop — AND — Omega bag cross corner loop

(both tables in the file are identical)

SWL	SF 5:1 fabric	SF 5:1 bottom	Lami	SF 6:1 fabric	SF 6:1 bottom	Lami
500	100 GSM with RF	100	20	110 with RF	110	20
501–750	125 with RF	125	25	130 with RF	130	25
751–1000	145–150 with RF	150	25	160 with RF	160	30
1000–1250	170 with RF	170	30	180 with RF	180	30
1251–1500	190 with RF	190	30	210 with RF	210	30
1501–1600	200 with RF	200	30	225 with RF	225	30–35
1601–1800	225 with RF	225	30–35	240 with RF	240	35
1801–2000	240 with RF	240	35	150×2 with RF (DOUBLE LAYER)	150×2	25

4-panel baffle (Q-bag) CROSS CORNER loop

SWL	SF 5:1	bottom	Lami	SF 6:1	bottom	Lami
500	120 with RF	120	20	110 with RF	110	20
501–750	135 with RF	135	25	130 with RF	130	25
751–1000	150 with RF	150	25	160 with RF	160	30
1000–1250	170 with RF	170	30	180 with RF	180	30
1251–1500	190 with RF	190	30	210 with RF	210	30
1501–1600	200 with RF	200	30	225 with RF	225	30–35
1601–1800	225 with RF	225	30–35	240 with RF	240	35
1801–2000	240 with RF	240	35	—	—	—

4-panel baffle (Q-bag) CORNER loop ← *the table an earlier analysis used for U-panel*

SWL	SF 5:1	bottom	Lami	SF 6:1	bottom	Lami
500	110 GSM	NA 110	20	120 GSM	NA 120	20
501–750	125	NA 125	20	135	NA 135	25
751–1000	**140**	NA 140	25	150	NA 150	25
1000–1250	**155–160**	NA 160	25	170	NA 170	30
1251–1500	180	180 NA	30	200	NA 200	30
1501–1600	190	~~NA~~ 190	30	220	NA 220	30
1601–1800	210	NA ~~210~~	30	230	NA 230	35
1801–2000	230–240	NA 230	35	—	—	—

Tunnel bag (corner loop)

SWL	SF 5:1	Lami	SF 6:1	Lami
500	100	20	110	20
501–750	120	20	130	25
751–1000	125–130	25	150	25
1000–1250	150–155	25	170	30
1251–1500	180	30	195–200	30
1501–1600	190	30	210–215	30
1601–1800	210	30	225	30–35
1801–2000	230–240	35	140–150 (DOUBLE LAYER)	25

Belt GPM — TWO PANEL / OMEGA / 4-PANEL BAFFLE (cross corner loop)

SWL	SF 5:1 width	SF 5:1 GPM	SF 6:1 width	SF 6:1 GPM
500	70–75 mm	35 g	70–75 mm	35 g
501–750	70–75 mm	35 g	75–80 mm	40 g
751–1000	75–80 mm	40 g	85–100 mm	45 g
1000–1250	78–100 mm	45 g	86–100 mm	50 g
1251–1500	79–100 mm	50 g	87–100 mm	55 g
1501–1600	80–100 mm	50–52 g	88–100 mm	55 g
1601–1800	81–100 mm	55 g	89–100 mm	60–65 g
1801–2000	82–100 mm	60–65 g	90–100 mm	50 g (DOUBLE LAYER)

Belt GPM — TUNNEL / BAFFLE / BOX (corner loop) ← *the table used for the 85×85×85 reference*

SWL	SF 5:1 width	SF 5:1 GPM	SF 6:1 width	SF 6:1 GPM
500	40–45 mm	22–25 g	40–45 mm	25–28 g
501–750	45–50 mm	25–28 g	50 mm	30–32 g
751–1000	**50 mm**	**30–35 g**	50 mm	40 g
1000–1250	50 mm	40 g	50 mm	45 g
1251–1500	50 mm	45–50 g	50 mm	50 g
1501–1600	50 mm	50 g	50 mm	55 g
1601–1800	50 mm	55 g	50 mm	60 g
1801–2000	50 mm	60 g	50 mm	40–45 g (DOUBLE LAYER AT FREE HEIGHT)

The 85×85×85 reference bag used **50 mm / 35 GPM** — the **top** of the 30–35 band. See question **C5**.

1.15 What the order actually supplies

Header field blank or "NA", across 395 sheets:

Field	Absent / NA	%
SWL	3	**0.8%**
bag weight	36	**9.1%**
packing	67	17.0%
printing	170	43.0%
doc pocket	176	44.6%
liner	310	**78.5%**

AS per Mentioned in PO

Family distribution across the 395 sheets: **u_panel 174 · circular 113 · baffle 70 · single_loop 14 · four_panel 11 · two_loop 10 · unclassified 133**

1.16 Sheet structure (all 395 follow this)

A1 ANJANI INTERWEAVE [PVT LTD.]
A2 BILL OF MATERIAL
A3/C3 CUSTOMER NAME
A4/C4 ORDER NO. (also spelt OEDER NO.)
A5/C5 REF NO.
A6/C6 SWL e.g. "750 KGS"
A7/C7 SF stored as Excel time 5:01 / 6:01
A8/C8 QTY
A9/C9 BAG TYPE free text, 60 variants
A10/C10 FABRIC COLOR
A11/C11 BAG WT. | REQ. WT. | REW. WT. <- the target weight
G3 PRINTING + H3 yes/no, I3 sides, J3 colours, K3 colour
G4 BAG SIZE + H4 I4 J4 = L W H + K4 "CM" + L4 "EXTERNAL"
G5 BAG SIZE + ... + L5 "INTERNAL"
G6 LOOP + H6 "4X25" + I6 "LL 60 CM" + J6 "SL 40 CM" + K6 colour + L6 "50 MM/WHITE"
G7 SKIRT TOP | TOP/FS + spec
G8 BOTTOM/DS + spec
G9 DOC PKT
G10 LINER
G11 PACKING | CONTAINER DATE

customer name As per PO mentioned.

A13:L31 COMPONENT TABLE, 15 fixed rows
SR NO. | COMPONENT | GSM | LAMI/UL | FABRIC COLOR | FABRIC WIDTH (CM) | CUT LENGTH(CM) | CUT PIECE WEIGHT | TOTAL | REMARK
rows: BODY PANEL, SIDE/BOTTOM PANNEL, TOP PANNEL, FILLING SPOUT, D/S, TIE/IRIS ROPE, FS TIE, DS TIE, LOOP, LINER, DDP, DOC, STITCHING, PRINTING, LABEL | Rope, B.tape, Glue, Tabbing tape. and extra component
closed by GRAND TOTAL

A31-A35 INSTRACTION :- free-text production lines
A36+ TAPE PLANT | LOOM DEPARTMENT | NIDDLE LOOM specs
A42 MIXING e.g. "PP - 80 % , CC - 8% , UV - 1.5 % , RP - 12.5 %"
A51 PREPARE BY / CHECKED BY / APPROVE BY

Rows are always listed, even when not fitted (weight = 0):

Row	Listed	Actually fitted	% fitted
BODY PANEL	394	394	100.0%
STITCHING	394	394	100.0%
LABEL	390	377	96.7%
LOOP	368	363	98.6%
SIDE PANNEL	152	148	97.4%
TOP PANNEL	189	131	69.3%
DOC	351	195	55.6%
D/S	187	98	52.4%
PRINTING	359	186	51.8%
FILLING SPOUT	184	95	51.6%
DS TIE	147	70	47.6%
LINER	176	72	40.9%
DDP	188	54	28.7%

1.17 Vocabulary found in the data

Bag type — 60 distinct raw strings (top 45)

54 U PANNEL WITH CORNER LOOP
46 U PANNEL BAG WITH CORNER LOOP
36 CIRCULAR WITH CROSS CORNER LOOP
25 4 PANNEL BAFFLE WITH CORNER LOOP
25 CIRCULAR BAG WITH CROSS CORNER LOOP
4 U PANNEL BAG
4 TUBELAR BAG WITH TOP HEMMING
4 U+2 PANNEL BAG WITH CORNER LOOP
4 CIRCULAR ROUND BAG WITH FULL RUNNING LOOP
4 DOUBLE LAYER U-PANNEL BAG

13 4 PANNEL BAFFLE BAG WITH CORNER LOOP 4 4 PANNEL BAG WITH CROSS CORNER LOOP
13 SINGLE LOOP BAG 4 U+2 WITH DOUBLE LAYER (STEVE' DOOR)
12 U+2 PANEL WITH CORNER LOOP 4 U+2 (140+25)GSM
11 U PANEL BAG 3 H BOTTOM BAG
10 CIRCULAR CROSS CORNER 3 2 LOOP BAG
10 U+2 3 CIRCULAR BAG WITH CROSS LOOP
10 CIRCULAR NAG WITH CROSS CORNER LOOP 3 4 PANNEL BAG WITH FLANGE LINER
9 U-PANNEL BAG 3 2 LOOP BAG WITH 4 CORNER LOOP
7 4 PANNEL BAFFLE BAG WITH CROSS CORNER 3 U PANNEL BAG CORNER LOOP
7 4 PANNEL BAFFLE BAG (Q BAG) 2 U+2 CORNER LOOP CONDUCTIV
6 4 PANNEL WITH BAFFLE BAG (Q BAG) 2 4-PANNEL WITH CORNER LOOP
5 CIRCULAR 2 CONTAINER LINER BAG TOP FULL OPEN WITH VELCRO AND BOTTOM SPOUT
4 CORNER LOOP + BAFFLE BAG 2 CIRCULAR BAG
4 4-CORNER LOOPBAFFLE BAG 2 4 PANNEL BAFFLE BAG
4 DOUBLE LOOP TUBELAR BAG 2 CIRCULAR BAG WITH OMEGA DESIGN
2 4 PANNEL WITH LINER BAG
2 U+2 PANNEL BAG
1 4 PANNEL WITH BAFFLE BAG
1 U+2 APNNEL BAG WITH DOUBLE LAYER
1 4 PANNEL BAFFLE WITH CROSS CORNER LOOP

Note the typos that still carry meaning: CIRCULAR NAG, 4-CORNER LOOPBAFFLE BAG (no space), U+2 APNNEL.

Component labels — 127 distinct (top 60)

394 BODY PANEL 152 SIDE PANNEL 37 SKIRT TIE 19 TOP / FS
394 STITCHING 147 DS TIE 37 DISCHARGE SPOUT 17 STEEVE DORE
390 LABEL 141 TIE/IRIS ROPE 31 F/S 17 TUCKING TIE
368 LOOP 110 BOTTOM PANNEL 31 LOOP PROTECTOR 17 STEEVEDORE
359 PRINTING 109 FS TIE 30 TIE/IRIS 16 EXTRA LOOP
351 DOC 98 BOTTOM PANEL 30 SIDE PANEL 16 TOP BEND
189 TOP PANNEL 79 IRISH TIE 25 TOP 15 FS & DS TIE
188 DDP 72 BAFFLE 25 DUFFLE 14 GLUE
187 D/S 71 TIE 25 FELT 14 F/S , D/S TIE
184 FILLING SPOUT 68 TOP PANEL 24 F/S & D/S TIE 13 FS/ DS TIE
176 LINER 59 D SPOUT 22 BRAIDED ROPE 12 DS/FS TIE

52 FS SPOUT	22 TABBING TAPE	10 SUPPORTING TIE
51 BAFFLE FABRIC	22 FILLERCORD	10 D/ SPOUT
51 SKIRT	21 BELLY BEND	9 DS SPOUT
48 TOP SPOUT	20 DS FLAP	8 TIE/DS
42 BOTTOM SPOUT	37 FS / DS TIE	8 BELLY BELT
	8 TOP/SKT	

Proposed canonical merges — need confirmation (question E1):

Canonical	Raw labels seen
DISCHARGE SPOUT	D/S · D SPOUT · DS SPOUT · DISCHARGE SPOUT · BOTTOM SPOUT · D/ SPOUT , outlet spout
FILLING SPOUT	FILLING SPOUT · FS SPOUT · F/S · TOP SPOUT · TOP / FS · TOP/~~SKT~~ TOP , inlet spout
SIDE PANEL	SIDE PANNEL · SIDE PANEL
TOP PANEL	TOP PANNEL · TOP PANEL
BOTTOM PANEL	BOTTOM PANNEL · BOTTOM PANEL / ~~bale~~ panel.
BAFFLE	BAFFLE · BAFFLE FABRIC
IRIS TIE	TIE/IRIS ROPE · IRISH TIE · TIE/IRIS · BRAIDED ROPE
FS TIE / DS TIE	FS TIE · DS TIE · TIE · FS/DS TIE · F/S & D/S TIE · FS & DS TIE · F/S , D/S TIE · DS/FS TIE · TIE/DS · FS/ DS TIE · SUPPORTING TIE · TUCKING TIE · SKIRT TIE
STEVEDORE LOOP	STEEVE DORE · STEEVEDORE
BELLY BAND	BELLY BEND · BELLY BELT
SKIRT	SKIRT · DUFFLE
FILLER CORD	FILLERCORD / ~~cord~~ dust proof
(unclassified)	FELT · GLUE · TABBING TAPE · TOP BEND · DS FLAP · LOOP · PROTECTOR · EXTRA LOOP Extra component

Production instruction lines — 127 distinct (top 45)

16 TOP AND BOTTOM ROUND DOUBLE CHAIN STITCHING WITH WHITE YARN
14 ALL SEAM DOUBLE DUST PROOF
12 U+2 PANEL BAG WITH J FOLD BOX STITCHING AND TOP OUT FOLD
12 LOOP TO BE PLACED OUTSIDE BLUE COLOR WITH POLYSTER YARN 10000 DNR
12 STITCHING DOUBLE CHAIN WITH WHITE YARN 3% UV TREATED MINIMUM YARN DNR 3200
12 PACKING SHOULD BE 50 PCS PER BALE WITH PALLET FUMIGATED PALLET TO USED
9 TOP SPOUT TIE TAPE OUTER FIXED AT 25 CM FROM TOP
9 U PANNEL BAG WITH CORNER LOOP
8 TOP AND BOTTOM ROUND SINGLE DOUBLE FOLD STITCHING WHITE YARN
8 COMPANY LABEL AND DOCUMENT POUCH WITH FRONT SIDE OPEN FABRIC DOC SAME COLORE AS BODY FABRIC
8 4 PANNEL BAG BAFFLE WITH CORNER LOOP
8 STITCHING OVER LOCK AND CHAIN STITCHING BLUE AND WHITE YARN
7 BOTTOM J FOLD AND TOP OUT FOLD STITCHING 3200 DNR WHITE YARN OVERLOCK + CHAIN
7 BELT 50 CM ATTECHED BY 6 PASS JUKKI MINIMUM 2000 KGF STRENGTH
7 TOP PERIMETER DOUBLE DUST PROOF AND REST ALL SEAM SINGLE DUST PROOF
7 BAG WT +/-2 %
6 ALL FOLD J- FOLD IN BODY
6 TOP FOLD OUT SIDE
6 SAFTY IN WHITE 2200 DNR AND OVER LOCK WITH 3600 DNR BLUE
6 STITCHING HR+DR WITH BLUE AND WHITE YARN
6 CIRCULAR BAG WITH TOP OUT FOLD AND BOTTOM J-FOLD STITCHING PP BLUE YARN HR+DR
6 4 BAFFLE WITH 80+20 GSM ONE SIDE LAMINATED 10 HOLES AND STITCHING FROM BOTOOM AND TOP 10 CM
6 CIRCULAR BAG WITH TOP AND BOTTOM J-FOLD STITCHING TOP AND TOP ROUND WITHOUT DDP AND BOTTOM AND ROUND DDP
6 BOTTOM OUT FOLD AND TOP OUT FOLD
6 U PANNEL BAG WITH CORNOR LOOP BLUE AND WHITE STRIPES
6 STITCHING OVERLOCK+CHAIN STITCHING BLUE YARN
6 TOP SPOUT DOUBLE CHAIN STITCHING
6 PACKING 200 PCS PER PALLET 48 PALLET 48X200 = 9600 PCS IN CONTAINER
5 LOOP OUT SIDE STITCHING WHITE COLOR
5 QUANTITY OF BUNDLE 60 BAG'S OF EACH.
5 ALL SEAM OUT SIDE FOLD BUTTERFLY
5 MAKE SURE THE BAG IN SIDE AND OUT SIDE CONTAMINATION FREE
4 CIRCULAR BAG WITH TOP AND BOTTOM J-FOLD STITCHING
4 JUKKI 6 PASS STITCHING WITH WHITE BLUE

4 COMPANY LABEL AND DOCUMENT POUCH WITH BACK SIDE FLAP RHS OPEN
4 LINER 200X360 CM 55 MICRON BOTH SIDE OPEN LOOSELY LINER FULL GUESTED
4 4 PANNEL BAFFLE BAG WITH TOP AND BOTTOM OT-FOLD STITCHING PP BLUE YARN HR+DR
4 COMPANY LABEL AND DOCUMENT POUCH WITH RHS OPEN
4 U+2 PANNEL BAG WITH DOUBLE LAYER BAG
4 BOX J-FOLD TOP OUT FOLD STITCHING BLUE AND WHITE YARN
4 BOTTOM J-FOLD AND TOP OUT FOLD
4 BOX J FOLD AND TOP OUT FOLD STITCHING
4 OPEN TOP AND FLAT BOTTOM WITH TOP HEMMING
4 HR AND DR STITCHING WITH BLUE AND WHITE YARN

Raw-material mixing recipes found

PP - 80 % , CC - 8% , UV - 1.5 % , RP - 12.5 %
PP - 80.5 % , CC - 8% , UV - 1.5 % , RP - 10 %
PP - 88.5 % , CC - 10% , UV - 1.5 % (from the 85x85x85 reference sheet)

1.18 Rules from published FIBC practice (external sources)

Cross-checked so the questions are asked in the industry's own terms. **Confirm these match Anjani practice.**

Topic	Published practice
Construction families	U-panel = one U-piece forms bottom + two sides, plus two panels. 4-panel = four pieces sewn to a square. Circular/tubular = woven as a tube, seams only top and bottom.
Shape retention	4-panel > U-panel > circular. Cost runs the other way.
Baffle / Q-bag	"Baffled", "formstable" and "Q-bag" all name the same thing — internal corner panels that stop the bag bulging.
Safety factor	5:1 = single-trip. 6:1 = multi-trip and UN-certified. Not a free choice — dictated by the customer's use case.
Governing	ISO 21898 (FIBC construction and testing). Also AS 3668.

Topic	Published practice
Type A	Plain woven PP, no static protection. Non-flammable contents only.
Type B	Low breakdown voltage, prevents propagating brush discharge. Dry flammable powders only, no flammable solvents or gases present.
Type C	Conductive threads interwoven. **Must be electrically grounded** during fill and discharge.
Type D	Static-dissipative fabric (e.g. CROHMIQ). No grounding required.
Manufacturing sequence	Tape extrusion (virgin PP + $CaCO_3$ + UV masterbatch, drawn to denier and tenacity) → weaving on flat or circular looms → separate needle-loom production of webbing → cutting → printing → stitching → finishing.
Denier	Grams per 9,000 m of yarn. So 3200 DNR is a thickness spec, not a weight.
Filler cord	PP rope or braided cord sewn into a seam to stop fine dust escaping — makes the bag sift-proof.
Hemming	Fold-and-sew (or glue) that stops cut fabric fraying and adds seam strength.
Liner types	Form-fit (fits the interior exactly, for contamination-critical), gusseted (pleated sides, larger volumes), lay-flat (unfolds during filling, suits automated lines).
Document pouch	PE or PP pouch for shipping/identifying documents. Also called pocket or envelope.